Q.1 In fuzzy implication, for any fuzzy number x, x.(η(x)) is equal to _____.

A. x2

B. x – x2

C. x

D. 1-x

Q.2 What is the primary characteristic of Nebla-Implications in fuzzy logic?

A. They use only the minimum operator.

B. They use only the maximum operator.

C. They are defined based on a continuous t-norm.

D. They are defined based on probabilistic interpretations.

Q.3 Let X={x1 , x2 , ..., xn } be a universal set and A, B are two fuzzy sets on X, the distance d(A, B) satisfy the condition:

A. -1≤d(A, B)≤0

B. 1≤d(A, B)

C. d(A, B)≤0

D. 0≤d(A, B)≤1

Q.4 Which of the following is true about Nebla-Implications?

A. They do not follow the law of the excluded middle.

B. They are a special case of classical implications.

C. They are defined using a residual operator.

D. They are only applicable in binary logic.

Q.5 Which of the following properties is NOT typically associated with Nebla-Implications?

A. Monotonicity

B. Continuity

C. Symmetry

D. Associativity

Q.6 If A(x, 0.41) and B(x, 0.49) are two fuzzy sets, then max(A, B)=------

A. 1

B. 0.49

C. 0

D. 0.41

Q.7 In the context of fuzzy sets, what does the Euclidean distance measure?

A. The total area under the fuzzy sets.

B. The sum of the differences between corresponding membership degrees.

C. The square root of the sum of squared differences between corresponding membership degrees.

D. The maximum difference between corresponding membership degrees.

Q.8 Which of the following statements about the distance between fuzzy sets is true?

A. It is always a positive value or zero.

B. It is only meaningful for triangular fuzzy sets.

C. It can only be calculated if the fuzzy sets have the same number of elements.

D. It always results in a crisp number.

Q.9 Nebla-Implications are often used in fuzzy logic because they:

A. Provide a way to handle partial truth values.

B. Can only be applied to Boolean variables.

C. Are simpler than classical implications.

D. Do not require t-norms.

Q.10 If A(x,0.1) and B(x,1.0) are two fuzzy sets, then max(A, B)=------

A. 0

B. 0.5

C. 0.1

D. 1
